

Skip-gram (Setup)

- Notation:
 - Denote the j -th wordform in the vocabulary with x_j
 - Denote the vocabulary size (number of distinct wordforms) with V
 - N is a hyperparameter that determines the dimensionality of the embeddings

Skip-gram (Model)

The probability of predicting the neighbor x_j given the word x_k

Soft-max over the network's output

$$P(x_j|x_k) = \frac{e^{w'[:,j] \cdot w[k,:]}}{\sum_{j'=1,\dots,V} e^{w'[:,j'] \cdot w[k,:]}}$$

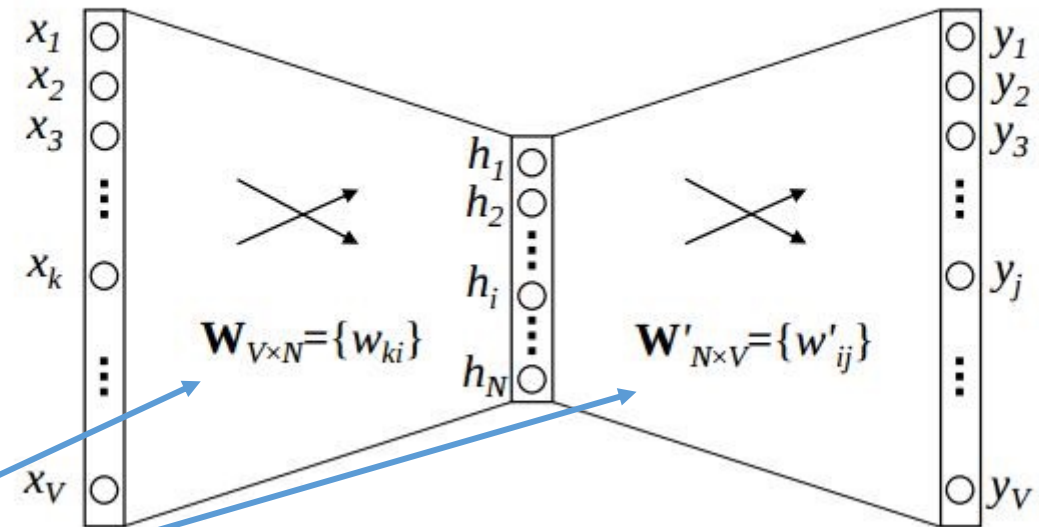
1-hot vector

Distribution over contexts

Input layer

Hidden layer

Output layer



W and W' are the parameters of the model

Skip-gram (Training)

- Training is carried out by maximizing

$$\operatorname{argmax}_{W, W'} \log[P(\text{text})] =$$

$$\operatorname{argmax}_{W, W'} \sum_{(x_j, x_k) \in \text{text}} \log[P(x_j | x_k)]$$

where x_k and x_j are any pair of words no more than K tokens apart

Skip-gram

Each wordform now has two embeddings:

input embedding in the input matrix W

- Row j of the input matrix W is the N dimensional embedding for word j in the vocabulary.

output embedding v' , in output matrix W'

- Column j of the output matrix W' is a N dimensional vector embedding for word j in the vocabulary.
- The input and output embeddings are often concatenated to yield the word embedding for j

